

Chemical Engineering FESP Mini-Project II: Fall 2018

Mini-Project II: Dye-Sensitized Solar Panels

In-class overview of solar panel technology and hands-on exercises where you will construct and make measurements on a dye-sensitized solar panel.

Project report: DUE FRIDAY NOVEMBER 30th

Using resources such as the overview by Dr. Behura, UIC library (books, reviews, journals, magazines), and the Internet, obtain information about solar panels and dye-sensitized solar panels. Obtain sufficient information to provide a full description of the application and how chemical engineering is involved in the technology. You may also include manufacturing, production and cost information if possible. In your report, include your findings from your in-class exercises including experimental procedures, experimental results, analysis and conclusions. Discuss your results and make conclusions regarding the data. Present all your findings in a report.

General project report format:

Title page

Abstract

Introduction/Background/Theory

Experimental Setup and Methods

Results and Discussion

Conclusions

References

Appendix of Raw Data (if needed)

Assignment Submission Instructions:

1. Save your report as a PDF file

Please use the following filename for your report file:

Dept_lastnameofpersonsubmitting_miniproject2.pdf

For example: CHE_Smith_miniproject2.pdf

2. One member of your team uploads the file to the Mini-Project II assignment submission link.

MINI-PROJECT II SCHEDULE:

Week 11: Tuesday November 6th or Thursday November 8th

- Dye-sensitized Solar Panel Overview by Dr. Sanjay Behura and Hisham Maddah
- Pick project teams

Week 12: Tuesday November 13th or Thursday November 15th

- Solar panel project: In-class exercises.
- Work on project report

Week 13: Tuesday November 20st or Thursday November 22rd

- Thanksgiving Week: NO CLASS TUESDAY and THURSDAY

Week 14: Tuesday November 28th or Thursday November 30th

- Project wrap-up, Project report due November 30th